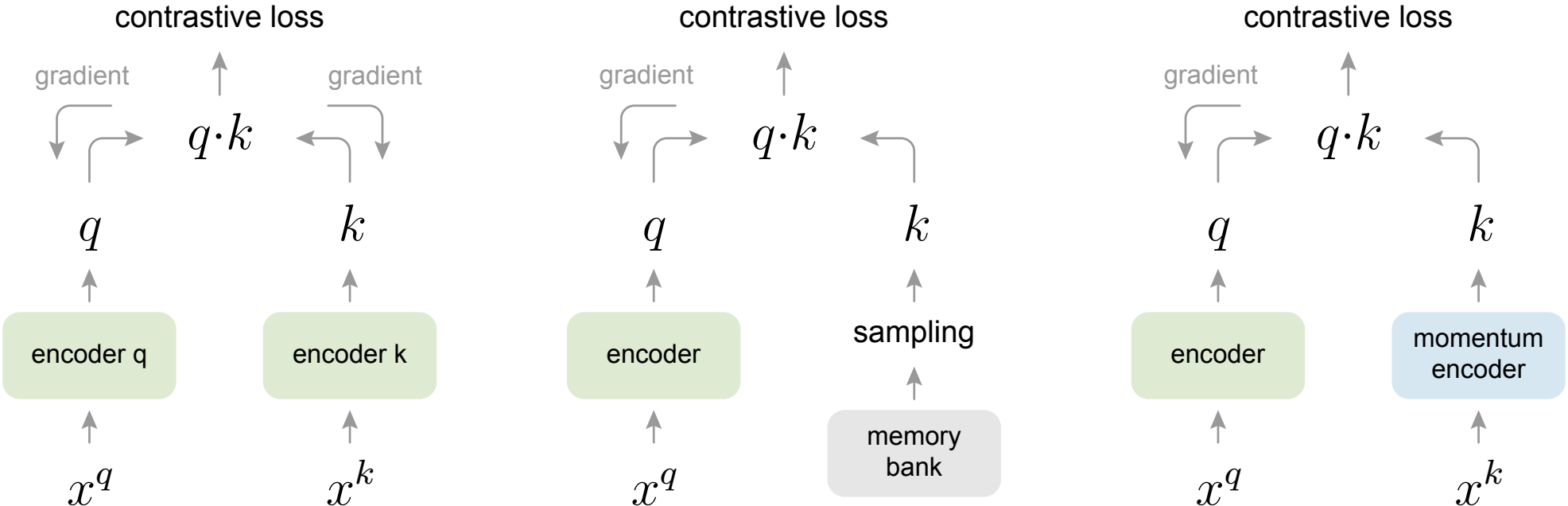

contrastive loss
gradient
gradient
$q \cdot k$
q
k
encoder q
encoder k
x^q
x^k
(a) end-to-end
contrastive loss
gradient
$q \cdot k$
q
k
encoder
sampling
memory bank
x^q
(b) memory bank
contrastive loss
gradient
$q \cdot k$
q
k
encoder
momentum encoder
x^q
x^k
(c) MoCo